Interactive Business Analytics Using R and Shiny

The Radiant Package

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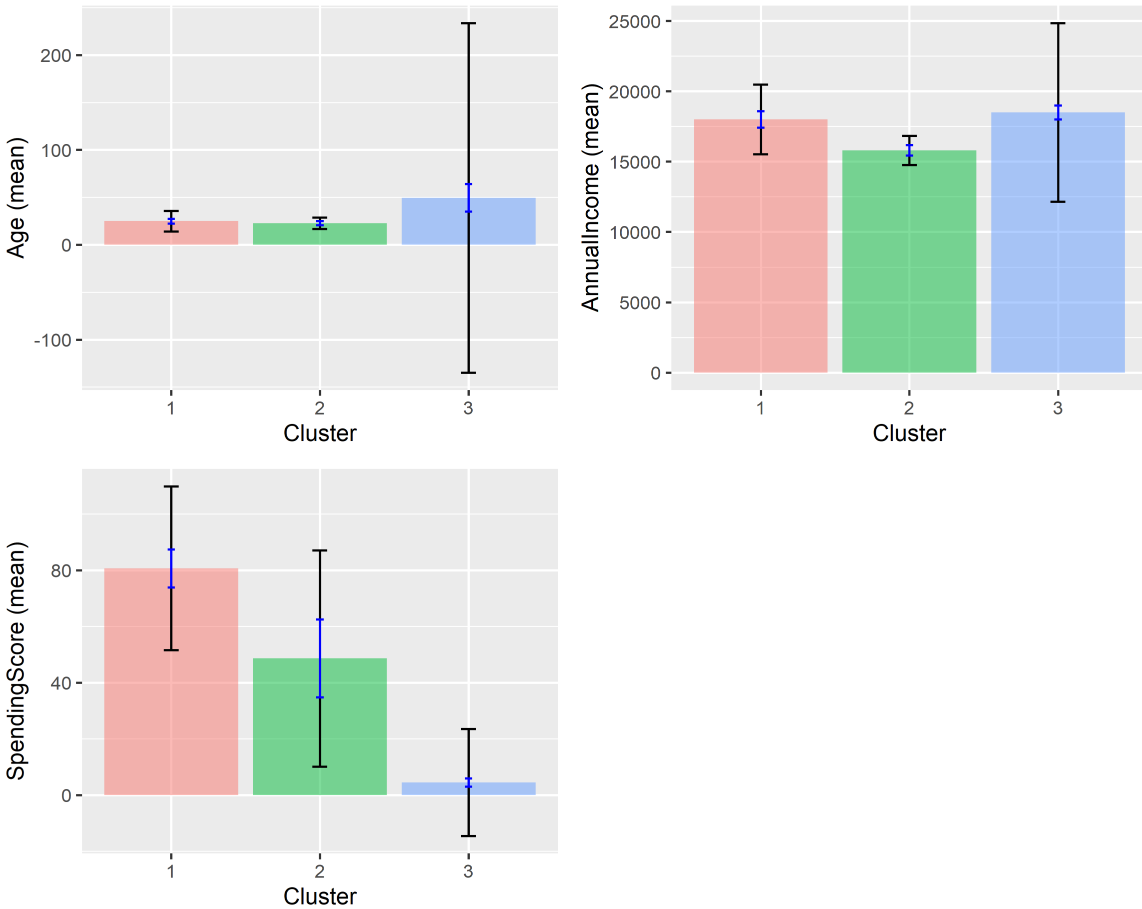
Interactive Business Analytics Dashboard



Presentation Outline

1. What is Interactive Business Analytics?
2. Traditional vs Interactive Analytics
3. Introduction to Radiant
4. The Shiny Framework
5. Live Demo: Customer Segmentation
6. Building Custom Apps & Summary

What is Interactive Business Analytics?

Definition: Dynamic tools that allow business users to explore data, change parameters, and see results in real time.

Real-Time Responsiveness

- Adjust a slider - plot updates instantly
- No waiting, no refresh button
- Immediate visual feedback

User-Driven Exploration

- Users control what they see
- Ask their own questions
- Discover patterns themselves

Accessibility

- Point-and-click interface
- No coding required
- Reduces dependency on analysts

Reproducibility

- All actions generate underlying R code
- Results can be verified and repeated
- Decisions are documented

Traditional vs Interactive Analytics

Feature	Traditional (Static)	Interactive (Dynamic)
Who explores?	Analyst creates report	User explores directly
Parameters	Fixed	Adjustable in real time
Turnaround	Days / hours	Instant
Insights	One-time	Continuous

Why does this matter?

- **Speed:** No waiting for analyst reports
- **Exploration:** Users answer their own questions

- **Scenarios:** Test multiple "what-if" situations
- **Democratisation:** Analytics for everyone

What is Radiant?

> Radiant is a platform-independent, browser-based interface for business analytics in R.

Key Features

- **Data:** Import, view, transform
- **Basics:** Explore, visualise
- **Model:** Regression, prediction
- **Multivariate:** Clustering, PCA
- **Report:** Export reproducible output

Installing and Launching

```r
install.packages("radiant")
library(radiant)
radiant()
```

Why Radiant?

- Built on R analytics + Shiny interactivity
- Point-and-click, no coding required
- Generates reproducible R code automatically
- Free and open-source
- Runs in any browser

The Shiny Framework

> Shiny is an R framework for building interactive web applications.

Core concept: Reactive Programming - outputs automatically update when inputs change.

```r
library(shiny)

ui <- fluidPage(
  sliderInput("threshold", "Sales Threshold:",
              min = 0, max = 100000, value = 50000),
  plotOutput("salesPlot")
)

server <- function(input, output) {
  output$salesPlot <- renderPlot({
    # Reacts automatically when input$threshold changes
  })
}

shinyApp(ui = ui, server = server)
```

- `ui` - controls what the user **sees**
- `server` - controls how the app **responds**
- `shinyApp()` - combines both parts

Demo: Customer Segmentation Setup

Business question: How can we segment customers by age, income, and spending behaviour?

Step 1 - Launch Radiant

```r
library(radiant)
radiant()
```

Step 2 - Load data: Data → Manage → Load data → select `customer_data.csv` → click **Load**

Step 3 - Verify dataset: Data → View → confirm columns are present:

Age	AnnualIncome	SpendingScore
19	15000	39
21	15000	81

Demo: Running the Analysis

Step 4-5 - Configure K-Means Clustering

Multivariate → Cluster → select variables: **Age, AnnualIncome, SpendingScore** → Method: **K-means** → k = **3** → click **Estimate**

Step 6 - Observe the interactive output

After running, Radiant displays:

- Cluster results table
- Cluster sizes and means
- Visual plots
- Summary statistics

> This is Shiny's reactivity in action; change k from 3 to 4 and all outputs update instantly.

Interpreting the Customer Segments

Segment 1 High-Value Customers

- High income
- High spending score

Strong purchasing power, already highly engaged. Priority for loyalty programmes.

Segment 2 Budget-Conscious Customers

- Lower income
- Lower/moderate spending

Price-sensitive group. Respond well to discounts and value-for-money offers.

Segment 3 Growth Potential Customers

- Good income
- Lower spending score

Ability to spend more but not yet engaged. Target with personalised campaigns.

Building Custom Shiny Apps

Why go beyond Radiant?

- Specific business workflows
- Company branding and databases
- Complex multi-step logic
- User-friendly interfaces for non-R users

Deployment Options

Platform	Best For	Cost
shinyapps.io	Quick prototypes and sharing	Free tier available
Shiny Server	Internal company use	Free (open source)
RStudio Connect	Enterprise deployment	Commercial

> Radiant gives you the interface. Shiny gives you the freedom to build your own.

Class Exercise

Question:

> Which R package would you use first to create an interactive business analytics dashboard: Radiant or Shiny and why?

Think about:

- Ease of use vs customisation
- Business purpose vs technical flexibility
- Who is the end user?
- What does your organisation already know?

There is no single correct answer. The goal is to justify your choice.

Summary

Key Takeaways

- Interactive analytics empowers users to explore data without coding
- **Radiant** = R + Shiny, packaged as a business analytics interface
- **Shiny** = the reactive framework that powers Radiant and custom apps
- Customer segmentation is a practical, high-value application
- Both tools are free, open-source, and run in the browser

The Core Relationship

```
# R
#    Shiny (reactive web framework)
#        Radiant (business analytics UI)
#            Your Custom Apps
```

Start with Radiant. Build with Shiny.

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Thank You